# Machine Learning for Spectral Analysis

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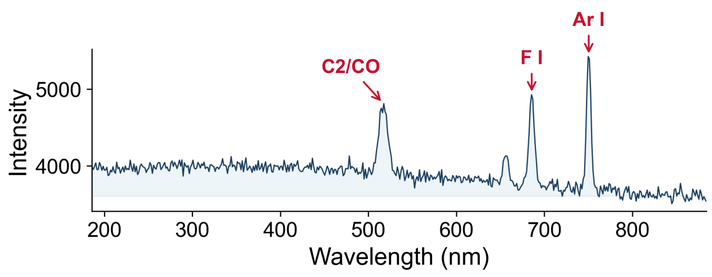
## 1. Introduction

*"How can we automate multi-species identification and plasma state classification from high-dimensional OES data?"*

### Core contributions:

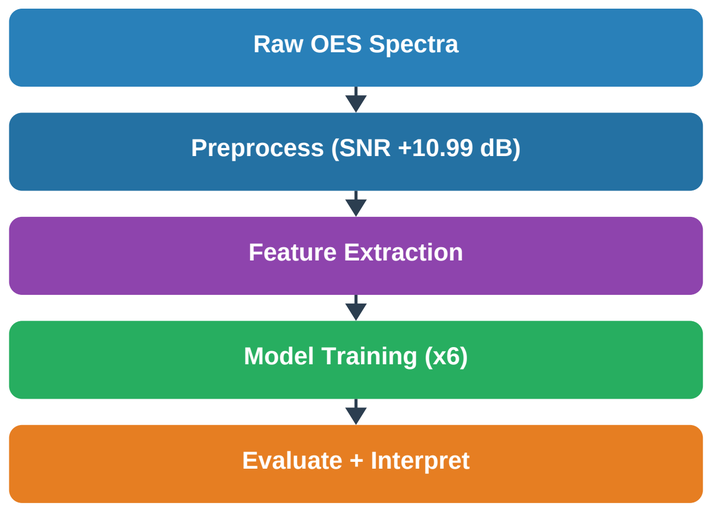
- **Species identification** via NMF + NIST matching
- **Plasma state classification** (6 models compared)

Also: actinometry, Boltzmann T_exc, temporal DTW clustering

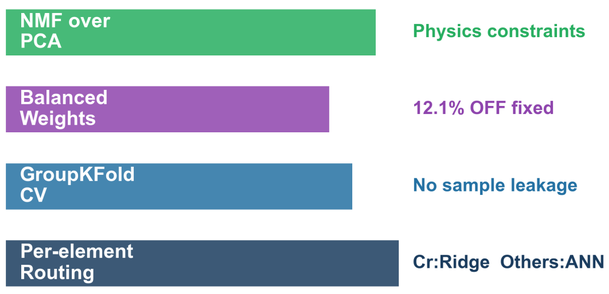



| Dataset | Channels | Task |
|---|---|---|
| BOSCH RIE | 3,648 | Plasma ON/OFF |
| Mesbah CAP | 51 | T_rot / T_vib |

## 2. Methodology

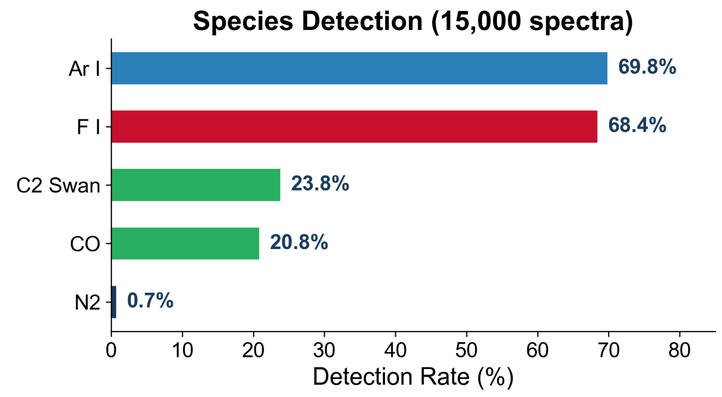



**Key Design Decisions**




## 3. Species Identification

**NMF (X = W . H)** decomposes mixed spectra into pure-species components.

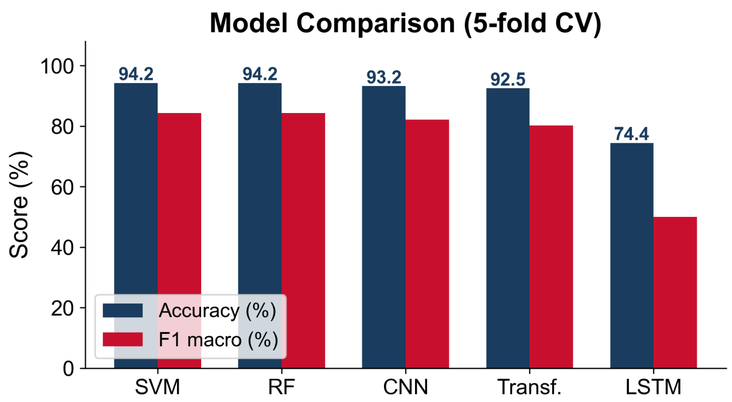



| Species | Lines (nm) | Origin |
|---|---|---|
| F I | 685.6, 703.7 | SF6 etchant |
| Ar I | 696.5, 750.4 | Carrier gas |
| C2 Swan | 516.5, 563.6 | C4F8 indicator |

**Validation:** NMF peaks at 684.4 nm (=F I) and 515.1 nm (=C2) confirm NIST database independently.

## 4. Classification Results

**F1 macro = 0.843 | Accuracy = 94.2%**

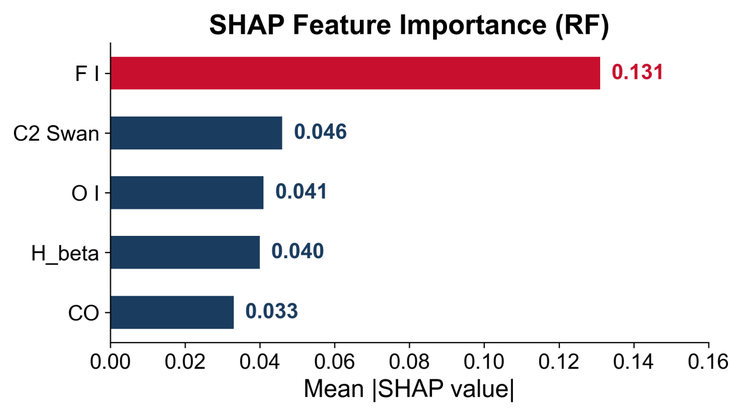



**Confusion Matrix (RF)**

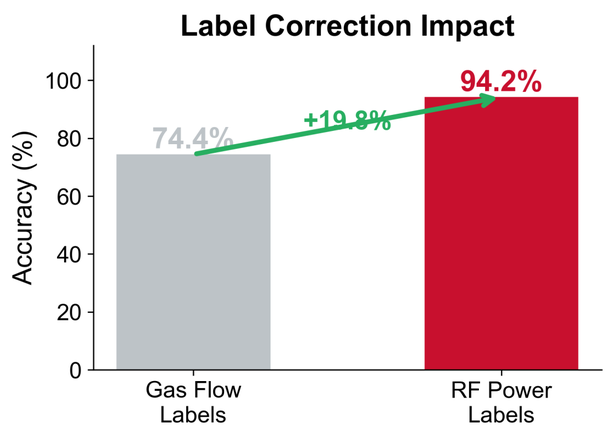
| | Pred OFF | Pred ON |
|---|---|---|
| True OFF | 1,105 | 706 |
| True ON | 132 | 10,057 |

**Note:** OFF recall=61% reflects 12.1% class imbalance. Balanced weights applied but minority detection remains challenging.

## 5. Interpretability & Physics




**SHAP confirms F I (fluorine radical) as the primary discriminator** — consistent with SF6 dissociation chemistry.




**Key discovery:** Initial gas-flow labels produced only 74% accuracy. Per-species intensity analysis revealed etch/passivation spectral difference < 0.35 std — indistinguishable. Switching to RF-power labels (plasma ON/OFF) improved to **94%**, proving **data quality, not model complexity, was the real bottleneck**.

**Boltzmann T_exc = 13,334 K** (6 Ar I lines, 696-772 nm).

## 6. Conclusions

**13 Species Detected**
39 NIST lines | NMF validated

**94.2% Accuracy | F1 = 0.843**
6 models compared (SVM/RF best)

**Label Correction: 74% -> 94%**
Data quality > model complexity

**SHAP Validates Physics**
F I = primary SF6 etchant radical

*Future: Multi-class tracking | Real-time streaming | Transfer learning*

### References:

[1] Gidon et al. (2019) IEEE Trans. Radiat. Plasma Med. Sci.
[2] Coburn & Chen (1980) J. Appl. Phys.
[3] Lee & Seung (1999) Nature — NMF
[4] Lundberg & Lee (2017) NeurIPS — SHAP
[5] Vaswani et al. (2017) NeurIPS — Transformer
[6] Contreras et al. (2024) Anal. Chem.
[7] BOSCH dataset: Zenodo #17122442